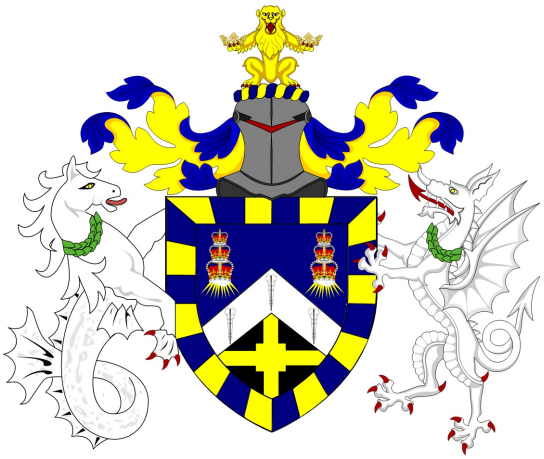
Data Analytics MSc Dissertation MTHM038, 2023/24

From Pitch to Price

A Machine Learning Framework for Analysing Player Performance and Market Values

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A thesis presented for the degree of Master of Science in *Data Analytics*

School of Mathematical Sciences

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Declaration of original work

This declaration is made on September 2, 2024.

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Abstract

This dissertation challenges the conventional reliance on regression-based methodologies for football player valuation, which often requires capturing the intricate dynamics of the transfer market over a period of time. Regression models frequently fail to address the volatility and noise inherent in market data, leading to imprecise predictions. Addressing these limitations, this study introduces a novel classification-based machine learning framework, leveraging performance metrics from FBref.com and market valuation data from Transfermarkt. The focus is categorizing player market values relative to the average, simplifying decision-making, and enhancing interpretability. By utilizing a custom voting classifier, the framework effectively navigates market data's stochastic nature, yielding more stable and actionable insights.

The research innovates with a sophisticated classification approach and meticulously curates a comprehensive dataset spanning Europe's top five leagues across two seasons. After the dissertation is evaluated, this dataset and position-specific subsets will be publicly available on Kaggle. The study's detailed feature importance analysis and model refinement contribute to a more accurate and equitable evaluation of player market values, with potential implications extending beyond football to other asset valuation contexts.

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Chapter 1

Introduction

1.1 Dissertation Overview

Many existing studies on football player valuation have primarily relied on regression-based machine learning models and basic statistical methods. These approaches may only partially capture the dynamic nature of the football transfer market. This complexity arises from the intricate relationships between player performance metrics, fluctuating market dynamics, and broader economic conditions. Ultimately, this can result in less precise and actionable insights.

This dissertation addresses some of these issues by employing a novel classification-based machine learning approach to predict football player market values. It focuses on players from Europe's top five leagues over the 2021-2022 and 2022-2023 seasons. By combining performance metrics from FBref.com [1] with the reliable market valuation data from Transfermarkt [2], this study constructs a comprehensive dataset to lay the foundation for a more impactful analysis of player market values.

The primary objective is to classify whether a player's market value is above or below the average. Additionally, this project aims to analyze key performance metrics —such as playing time, pass accuracy, and league par-

ticipation—to identify the most influential factors affecting player valuations. A thorough feature importance analysis provided insights into the key variables influencing player market values. To implement our data analysis pipeline, various machine learning models, statistical techniques, and evaluation methodologies were utilized to tackle various challenges.
A range of machine learning models were implemented and evaluated using accuracy and F1 scores. After analyzing individual models, a custom voting classifier specific to our data model was also used to improve predictive performance. Overall, this framework can have implications beyond football, extending to other sports and industries where asset valuation is essential.

1.2 Rules of Football

A game of football is played between two teams, each comprising eleven players. The main objective is to score more goals than the opposing team by maneuvering the ball into their net. The game is played over two halves, each lasting 45 minutes, with a brief halftime interval[3]. Depending on the league or competition in which the match is played, there is the option to substitute a fixed number of players from a designated bench. However, there is no obligation to make these substitutions. Players can use any part of their body except their hands and arms to control the ball. The only exception is the goalkeeper, who is permitted to use their hands within a designated penalty area [4].

1.3 Player Positioning and Tactical Responsibilities

In football, players are assigned positions that come with distinct duties that they must fulfill during a match. The goalkeeper's primary role involves guarding the goalpost and preventing the opposing team from scoring goals

by using their hands and body within the penalty area. Defenders position themselves near their goal to stop the opposing team's attempts at scoring. They do this by intercepting passes and blocking shots. Midfielders operate in the center part of the field, contributing to both defensive and offensive play. They facilitate ball movement through passing to create goal-scoring opportunities. Midfielders are considered to dictate the pace and flow of the game. Similarly, attackers focus on scoring goals. They play near the opponent's goal and try to take shots or create opportunities for others to score [4]. Additionally, these roles can have variations based on positioning on the pitch. For example, midfielders can be categorized as left or right midfielders, central attacking midfielders, and central defensive midfielders. Defenders and attackers also have such variations. These include positions like left-back, right-back, and center-back for defenders and left-wing, right-wing, and center-forward for attackers.
Some common positions include:

- Goalkeepers: Goalkeeper (GK)
- Defenders: Left-back (LB), Center-back (CB), Right-back (RB), Left Center-back (LCB), Right Center-back (RCB)
- Midfielders: Central Defensive Midfielder (CDM), Left Midfielder (LM), Right Midfielder (RM), Center Midfielder (CM), Central Attacking Midfielder (CAM)
- Attackers: Left Winger (LW), Right Winger (RW), Center Forward (CF), Striker (ST)

The diagram below (Figure 1.1) illustrates typical football player roles and their positioning on the pitch.

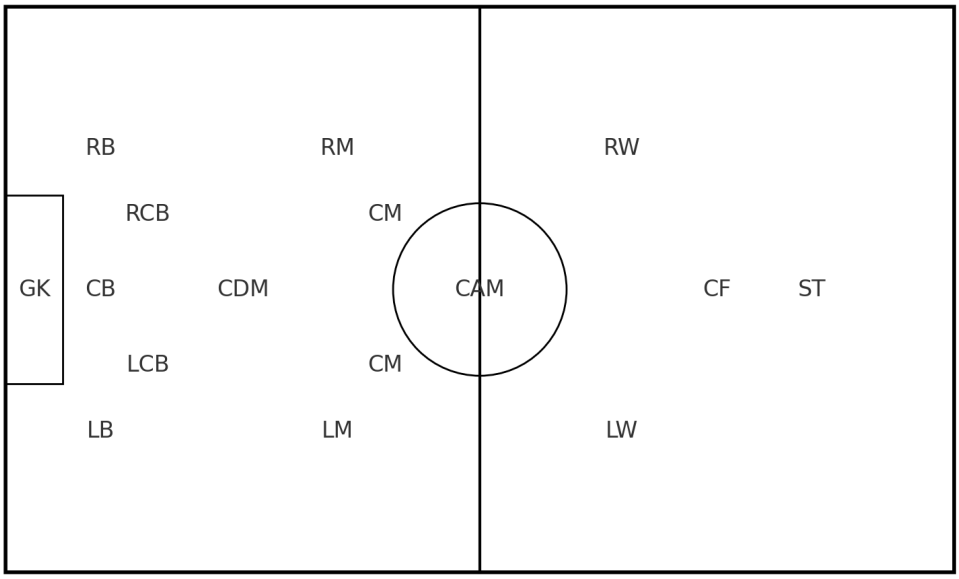


Figure 1.1: : An illustration of typical football player roles and their positions in a standard formation

.

1.4 Key Events in a Football Game

In a game of football, certain key events play a pivotal role in determining the outcome of a match. These events, which range from goals scored to defensive actions, are meticulously recorded and analyzed by data collection platforms. Clubs and analysts utilize this data to make informed decisions, whether in player transfers, in-game tactics, or long-term strategic planning. By examining these events at a granular level, clubs can predict match outcomes, evaluate player performance, and optimize their overall strategy. Thus, these events not only shape the dynamics of the game but also serve as critical features in predictive models and decision-making processes[5]. The following categorization of metrics outlines some of the most analyzed events in football and explains their significance in evaluating teams and players:

1.4.1 Attacking Metrics

Scoring a *goal* is the most critical event in a football match, which directly determines the match's outcome. A goal is scored when the ball is successfully kicked into the opposing team's net. Similarly, an *assist* in football is achieved by a player when the player passes the ball to a teammate who then scores a goal. The timing and frequency of goals can provide insights into a team's offensive capabilities and match strategy, making this data a vital feature for machine learning models that predict match results and player contributions [6].

Shots on Target encompass both goals scored and shots that require a save from the goalkeeper. Shots on target are key indicators of a team's attacking capability and efficiency. A high number of shots on target often indicates effective offensive play and pressure on the opposing team. Analyzing these metrics provides insight into a team's ability to convert opportunities into goals. Research has shown that the quality and quantity of shots on target correlate with successful attacking play and overall team performance[7].

Set Pieces include corners, free-kicks, and penalty kicks, which are awarded when play is halted due to fouls or the ball going out of bounds. Set pieces are crucial for scoring opportunities, especially in closely contested matches. Analyzing the effectiveness of set pieces offers insights into a team's tactical approach and ability to capitalize on these fixed chances. The strategic importance of set pieces and their impact on scoring opportunities is well-documented[6].

1.4.2 Possession and Passing Metrics

Possession refers to the percentage of time a team controls the ball during a match. Teams with higher possession rates often dominate play, create more scoring opportunities, and control the pace of the play. Possession is a significant factor in predicting match outcomes, as it correlates with both

offensive and defensive control. Studies have shown that possession percentage has a significant correlation with match results and shapes a team's offensive and defensive strategies[8]. Additionally, *Pass Completion Rate* evaluates the accuracy of passes between players on the same team. A high pass completion rate indicates a team's ability to maintain possession and organizing attacks. It demonstrates tactical discipline and the capability to execute complex game plans. Teams with high pass accuracy are often more successful as they can keep the ball away from their opponents and create scoring opportunities. Studies show a strong connection between high pass completion rates and team success[8].

1.4.3 Defensive Metrics

Defensive actions are crucial for preventing the opposing team from scoring. Key defensive maneuvers include *tackles*, *interceptions*, and clearances. Tackles involve a player challenging an opponent to win possession of the ball, whether it be a standing tackle (where the player remains on their feet) or a sliding tackle (where the player slides on the ground to reach the ball). *Interceptions* occur when a player anticipates an opponent's pass and steps in to gain possession. *Clearances* involve kicking or heading the ball away from the goal area to prevent scoring. These defensive metrics are crucial to assess a team's ability to disrupt opposition attacks and maintain defensive stability[9].

Fouls refer to actions that violate the rules of the game, such as tripping or pushing an opponent. Committing a higher count of fouls can disrupt the flow of the game and may lead to penalties or player suspensions. Fouls received can indicate how often a team is targeted by aggressive play, which could result in free-kick opportunities and alter match dynamics[10].

Chapter 2

Literature Review

2.1 The Role of Data Analytics in Football

Data analytics has revolutionized decision-making in football. It has enabled a shift from subjective judgments to objective and data-driven strategies. Within this domain, predictive modeling has emerged as a potent tool, allowing clubs, scouts, and analysts to derive actionable insights from vast datasets. These insights extend across various applications. These include but are not limited to player performance evaluation, tactical analysis, and injury prevention. Notably, the application of machine learning techniques in predicting player market values has gained significant attention. For instance, research illustrates how predictive models based on player statistics and match data can accurately forecast market values and future performance outcomes [11].

The increasing reliance on data analytics diverges from traditional, experience-based decision-making in football. Instead, teams operate within a data-driven framework that enhances decision-making processes. Additionally, data analytics can play a role in injury management. The Internet of Things (IoT) and Electronic Performance and Tracking Systems (EPTS) can collect and analyze data to predict and prevent injuries. This proactive approach re-

duces the risk of injury and prolongs player careers, ultimately contributing to better team performance [12]. Collectively, these applications highlight the transformative impact of data analytics and machine learning in modern football.

2.2 Predictive Models for Player Market Valuation

At the heart of football analytics lies the challenge of accurately predicting player market values. This task involves developing sophisticated models that can capture the intricate dynamics of the football transfer market. Recent advances in machine learning have introduced techniques such as clustering and hybrid models that offer promising results. For instance, advanced clustering techniques enable the grouping of players based on performance metrics, allowing the creation of tailored predictive models that account for the unique contributions of different player roles. Studies have demonstrated the efficacy of these clustering techniques in quantifying player performance and categorizing roles. Overall, this has enhanced the precision of market value predictions [13].

Beyond traditional performance metrics, incorporating diverse data sources, such as football video game data, has become increasingly common in the player market valuation domain. Laros G. explores the predictability of transfer values using machine learning techniques applied to FIFA video game data. This approach demonstrates how video game data closely simulates real-world conditions and can complement traditional data sources and enhance prediction accuracy [14]. Another study expands on this further by integrating particle swarm optimization with support vector regression using FIFA 20 game data. Particle Swarm Optimisation is an optimization algorithm inspired by the social behavior of birds and fish. It is used to find the optimal solutions in a given problem space by simulating the movement of

a group of particles. The study makes use of PSO for feature selection and parameter tuning. This methodology, which also underscores the importance of position-specific skills, aligns with and expands upon Laros's findings [15]. Another potential data source for football analytics is crowd-sourced estimates, mainly from platforms like Transfermarkt.com. These valuations rely on the collective intelligence of the user base, providing a unique perspective on market values. These inaccuracies orginate due to inexperienced members, a bias towards well-known players, and infrequent updates leading to unreliable market value estimations [15]. In contrast, Frick B. and Prockl F. delve into the reliability of these crowd-sourced valuations, concluding that they serve as reliable proxies for player salaries and valuations. The study demonstrates and advocates for the potential of Trasnfermarkt.com data to inform and enhance valuation models [16].

2.3 Gaps in Existing Literature

Despite advancements in player market valuation models, significant gaps remain, demanding further research to enhance predictive accuracy and robustness. Many existing models rely on a narrow set of performance metrics, such as passes completed or shots on target, limiting their ability to capture the full spectrum of factors influencing market value. Compounding this issue is the prevalent use of regression-based methods. While helpful in deriving numerical estimates, they fail to address market data's volatility and noise holistically. This reliance results in less reliable and actionable outcomes. As Behravan I. and Razavi S.M. mentioned, this narrow focus needs to be more balanced with the complexities of the football transfer market. They propose a hybrid machine learning approach that combines clustering with regression [15].

2.4 Conclusion

While the existing literature on football player market valuation offers various innovative approaches, several significant gaps remain. Most notably, current models heavily rely on regression-based methods and a limited range of performance metrics, which may not fully capture the complex and dynamic nature of the football transfer market. These methods often struggle to handle market data's volatility, noise, and outlier effects. Consequently, this leads to predictions that may not be accurate and thus actionable. Additionally, there is insufficient integration of diverse data types, which restricts the development of more comprehensive and adaptable valuation frameworks.
This dissertation addresses these gaps by introducing a novel classification-based approach to predict player market values by diverging from the traditional regression focus. The study aims to simplify the decision-making process and enhance Interpretability by categorizing market values instead of estimating precise numerical figures. The broader objective is to make the findings more practical for football clubs and scouts. Furthermore, classification models provide resistance against the noise and volatility in market data and enable more stable and reliable predictions. This method could improve model performance and scalability and potentially extend the approach's applicability to other industries where asset valuation is critical. By leveraging a unique dataset that integrates performance metrics with market valuation data and employing various machine learning techniques, this study provides a more comprehensive framework for player valuation. Including diverse data sources and advanced classification techniques offers a new perspective on handling the complexities of the football transfer market. Ultimately, we aim to contribute to more accurate and actionable insights for stakeholders.

Chapter 3

Datasets Overview

3.1 Introduction

In this dissertation, we leveraged two distinct yet complementary data sources: the Transfermarkt dataset and the FBref player statistics dataset. The Transfermarkt dataset primarily served as the foundation for player market valuations, while the FBref dataset provided detailed, granular performance statistics, including per-90-minute metrics, for players competing in Europe's top five leagues during the 2021-2022 and 2022-2023 seasons. Through the integration of these datasets, we constructed a comprehensive dataset that meticulously aligns individual player performance metrics with their corresponding market valuations across two different seasons. This approach allowed for a more nuanced analysis of the factors that drive market value fluctuations. In the subsequent sections, we offer a thorough exposition of these datasets, contextualizing their relevance and application within the scope of our dissertation.

3.2 Football Data from Transfermarkt

3.2.1 Description

This dataset captures football-related metrics, including player scores, valuations, transfer histories, match statistics, and other performance indicators from 14 leagues worldwide.
The dataset comprises 8 tables, each detailing different aspects of football players or clubs. :

1. **Competitions**: Tracks different football competitions using a unique `competition_id`.
2. **Games**: Records each match with a unique `game_id`, including participating clubs and associated competition details.
3. **Clubs**: Provides club information linked with a `club_id`, facilitating analysis of club-specific data.
4. **Players**: Contains player profiles, each associated with a `player_id`, enabling individual performance tracking.
5. **Appearances**: Logs each player's performance per game, with metrics such as goals, assists, and disciplinary records.
6. **Club Games**: Acts as a junction table linking clubs to specific games.
7. **Game Events**: Documents in-game events like goals, substitutions, and card incidents.
8. **Player Valuations**: Includes financial data related to players, allowing for market valuation analysis.

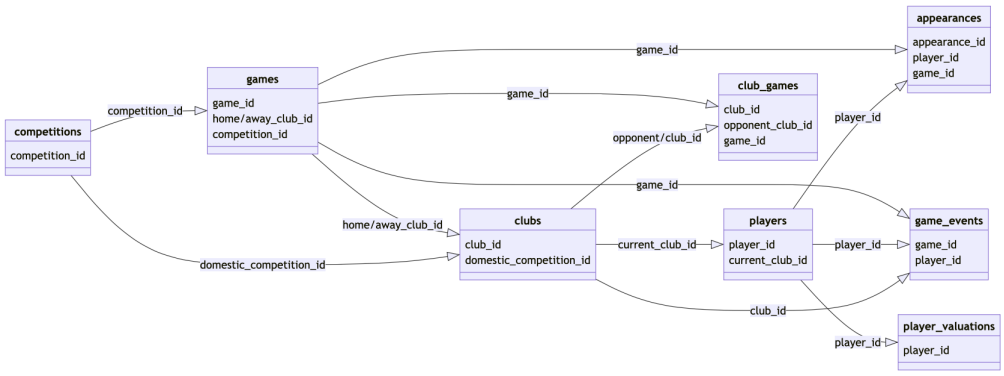


Figure 3.1: Dataset Model of the Football Data from Transfermarkt. Retrieved from: Football Data from Transfermarkt kaggle page **[3]**

.

The dataset's structure, as shown in 3.1, is by nature relational. It utilises primary keys like `player_id`, `game_id`, `club_id`, and `competition_id` to interconnect various tables. This design offers analytical advantages, such as linking different metrics and executing holistic analyses.

This dataset is scrapped from Transkermrkt website and maintained on Kaggle. This dataset updates weekly on Kaggle through an automated data pipeline that performs data scraping, transformation, and storage. This process ensures that the dataset remains current and reflects ongoing developments in the football world. Details about the source code and the pipeline process can be found in the Kaggle description and the associated GitHub repository [2].

3.2.2 Market valuations

Given that our dissertation is related to market value analysis, it is essential to note that Transfermarkt.de generates football valuation data through crowd wisdom. The platform allows users to individually assess player market values, which are then aggregated by experienced community members

and called "judges." This selective aggregation process helps filter outliers and ensures valuations reflect cumulative and informed opinions.
Notably, the study "Information precision in online communities: player valuations on www.transfermarkt.de" has validated this approach. It evaluates a strong correlation between these market values and actual player salaries, making them valuable for empirical research in sports economics. The findings indicate that these market values are excellent proxies for undisclosed player salaries, often confidential in many leagues. Consequently, the paper encourages researchers to utilize data from Transfermarkt.de in empirical studies related to sports economics, highlighting its reliability and relevance in analyzing transfer market metrics such as transfer values and probabilities compared to other options [16].

3.3 Footballref Player Performance Statistics datasets (2021-2023)

3.3.1 Description

The 2021-2022 [17] and 2022-2023 [18] Football Player Statistics datasets, both sourced from the Football Reference website [1] and uploaded to Kaggle, provide detailed statistics covering the top five European football leagues. These comprise the Premier League, Ligue 1, Bundesliga, Serie A, and La Liga. The Footballref website has also been used as a source for football data in previous academic research. For instance, Malagón Selma used data from fbref.com to develop machine learning models and employ multivariate statistical tools for football analytics [19]. Both datasets include player statistics calculated per 90 minutes covering either 36 or 38 league games based on the league or country in question and contain statistics for over 2500 players each. However, the 2021-2022 season dataset has more features (143) compared to the 2022-2023 season dataset (124). While merging the

dataset, we had to consider this aspect and merge the datasets across two seasons accordingly.

3.3.2 Key Metrics and Categories

To enhance the clarity and understanding of the data, the metrics from both datasets were categorized based on their nature and purpose. This self-devised categorization aims to make it easier to interpret and analyze the key metrics within the context of this dissertation.

Table 3.1 presents some of the key metrics from the 2022-2023 Football Player Statistics dataset [18] and the 2021-2022 Football Player Statistics dataset [17] after being categorized. The metrics are grouped into nine main categories, described as follows:

1. **Player Profile Information**: Includes the player's name, nationality, position, team (squad), and age.

2. **Playing Time**: Covers metrics such as matches played, matches started, total minutes played, and minutes played per 90 minutes.

3. **Offensive Contributions**: Provides data on goals scored, total shots, shots on target, shooting accuracy, goals per shot, and penalty kick statistics.

4. **Passing Statistics**: Includes passes completed, passes attempted, pass completion percentage, total pass distance, progressive passes, key passes, and assists.

5. **Defensive Statistics**: Encompasses tackles made, tackles won, interceptions, clearances, errors leading to opponent shots, blocks, pressures applied, and defensive actions that result in shots or goals.

6. **Dribbling Statistics**: Details dribbles attempted, dribbles completed successfully, dribbling success rate, and the number of opponents dribbled past.

7. **Carrying Statistics**: Covers the number of times the ball was carried, total carry distance, progressive carries, and carries into the penalty area.

8. **Reception Metrics**: Includes the number of times a player was the target of a pass, successfully received a pass, and the percentage of successful receptions.

9. **Disciplinary Statistics**: Provides information on fouls committed, fouls drawn, yellow and red cards, second yellow cards, offsides, own goals, and aerial duels won.

Detailed descriptions of each feature, outlining the meaning and application of every metric, can be found on Kaggle [18],[17].

Type of Variables	**Metrics and Abbreviations**
Player Profile Information	Player's Name (Player), Nationality (Nation), Position (Pos), Team (Squad), Age (Age)
Playing Time	Matches Played (MP), Matches Started (Starts), Total Minutes Played (Min), Minutes Played per 90 Minutes (90s)
Offensive Contributions	Goals Scored (Goals), Total Shots (Shots), Shots on Target (SoT), Shooting Accuracy (SoT%), Goals per Shot (G/Sh), Penalty Kick Goals (ShoPK), Penalty Kicks Attempted (PKatt), Penalty Kicks Won (PKwon), Shots from Free Kicks (ShoFK), Goal-Creating Actions (GCA): Goal-Creating Actions from Passes (GcaPassLive), Shots Leading to Goals (GcaSh), Fouls Drawn Leading to Goals (GcaFld), Defensive Efforts Leading to Goals (GcaDef), Dribbles Leading to Goals (GcaDrib), Shot-Creating Actions (SCA): Shots Created that Lead to Another Shot (ScaSh), Passes Leading to Shots (ScaPassLive), Defensive Actions Leading to Shots (ScaDef), Successful Dribbles Leading to Shots (ScaDrib), Fouls Drawn Leading to Shots (ScaFld)

Type of Variables	Key Metrics and Abbreviations
Passing Statistics	Passes Completed (PasTotCmp), Passes Attempted (PasTotAtt), Pass Completion Percentage (PasTotCmp%), Total Pass Distance (PasTotDist), Progressive Passes (PasProg), Key Passes (KP), Assists (Assists), Passes Between Defenders into Open Space (TB), Long Pass Completion Percentage (PasLonCmp%), Completed Passes into Final Third (Pas3rd), Passes that Lead to Goal (GcaPassDead), Passes Under Pressure (PasPress)
Defensive Statistics	Tackles Made (Tkl), Tackles Won (TklWon), Interceptions (Int), Clearances (Clr), Errors Leading to Opponent Shots (Err), Blocks (Blocks), Pressures Applied (Press), Defensive Actions Leading to Shots/Goals (ScaDef, GcaDef), Penalty Kicks Conceded (PKcon)
Dribbling Statistics	Dribbles Attempted (DriAtt), Dribbles Completed Successfully (DriSucc), Dribbling Success Rate (DriSucc%), Opponents Dribbled Past (DriPast), Successful Dribbles Leading to Shot Attempts (ScaDrib), Goal-Creating Dribbles (GcaDrib)
Carrying Statistics	Number of Ball Carries (Carries), Total Carry Distance (CarTotDist), Progressive Carries (CarProg), Carries into Penalty Area (CPA), Carries into Final Third (Car3rd)
Reception Metrics	Times Targeted by Pass (RecTarg), Successful Receptions (Rec), Successful Reception Percentage (Rec%), Passes Received Progressively (RecProg)
Disciplinary Statistics	Fouls Committed (Fls), Fouls Drawn (Fld), Yellow Cards (CrdY), Red Cards (CrdR), Second Yellow Cards (2CrdY), Offsides (Off), Own Goals (OG), Aerial Duels Won (AerWon)

Table 3.1: Key metrics from the 2022-2023 and 2021-2022 Football Player Statistics datasets, categorized to enhance readability and understanding.

Chapter 4

Methodology

4.1 Overview and Objectives

This study utilizes machine learning techniques to predict football player market values by analyzing performance metrics from the 2021-2022 [17] and 2022-2023 [18] seasons. The analysis combines player statistics from Footballref.com with market valuations from Transfermarkt [16], chosen for their comprehensive and reliable football data. The primary objective is to develop models that predict whether a player's market value is above or below the average for squads in Europe's top five leagues. Additionally, the study aims to identify key performance metrics, such as playing time, pass accuracy, and league participation, that significantly influence a player's market value. Unlike traditional regression-based approaches, this research adopts a classification-based method to predict player market values, offering a more straightforward and more actionable framework for decision-makers in football. This approach addresses a gap in the existing literature by providing a more accessible means of player valuation that could extend beyond football to other sports and industries where asset valuation is critical. Benefits of the classification approach include:

- **Simplification and Interpretability**: By predicting whether a player's

market value is above or below the average, the problem is simplified, making the results more interpretable and actionable for decision-makers.

- **Robustness Against Noise**: Market values can be volatile and influenced by various external factors. Binning these values into categories mitigates the impact of noise and reduces the influence of outliers, leading to more stable and reliable predictions.
- **Practical Decision-Making**: The binary nature of classification aligns well with real-world decision-making processes, where clubs and scouts often need to make clear-cut choices rather than relying on precise but potentially uncertain numerical estimates.
- **Model Performance and Feasibility**: Classification models are generally easier to implement and tune, and they offer clear evaluation metrics like accuracy and F1 score, which are more meaningful in binary decision contexts.
- **Scalability and Flexibility**: The classification approach can easily be adapted to different contexts or expanded into multi-class problems if needed, providing flexibility in applying the model.

By focusing on these benefits, the study aims to develop a robust and practical model for predicting player market values that enhance decision-making for football clubs, scouts, and potentially other fields requiring asset valuation.

4.2 Data Preparation and Analysis

The dataset was created by merging player performance metrics from Footballref for the 2021-2022 and 2022-2023 seasons with market valuations from Transfermarkt. The market values showed a highly skewed distribution, with

most players valued below the mean. This distribution led to a natural split: 2,284 players were classified as below average, while 858 were classified as above average. Data from each season was analyzed separately before being merged into a final dataset to account for season-specific factors such as market trends and economic conditions.

Given the inherent class imbalance, our analysis focused on identifying key performance metrics differentiating higher-valued players. To understand their relationships, we began with a correlation analysis of the numerical features against the raw market values. The analysis informed our binning process, ensuring the categorization was based on solid correlations and accurately reflected the factors influencing player valuations.

4.3 Model Selection and Evaluation

Various machine learning techniques, including Random Forest, Gradient Boosting, Logistic Regression, and Support Vector Machine classifiers, were selected due to their effectiveness in classification tasks. These models were evaluated using accuracy and F1 score, with ensemble methods like the Voting Classifier incorporated to assess whether combining individual models could improve predictive performance. To ensure robustness, extensive hyperparameter optimization was conducted using GridSearch, and stratified K-fold cross-validation was employed to optimize model performance and minimize overfitting.

In addition to the correlation analysis, after training the models, we conducted a detailed examination of the importance of features for models that support it, such as Random Forest and Gradient Boosting. This analysis helped identify which performance metrics most significantly affect a player's market value. We also analyzed the Logistic Regression model coefficients to assess each feature's influence. This comprehensive approach to feature importance provided valuable insights into which factors are most influential

in determining a player's market worth.
Finally, we create a custom voting classifier based on our insights from the feature importance analysis and model results.

4.4 Computational Framework

The Python programming language in a Jupyter Notebooks environment was employed due to its extensive library support and ease of use in data processing and machine learning model training. Key Python libraries used included:

- **Numpy:** For numerical computations.
- **Pandas:** For data manipulation, merging, filtering, and cleaning.
- **Matplotlib and Seaborn:** For data visualization.
- **Scikit-learn:** For implementing machine learning models and evaluation techniques.
- **XGBoost and CatBoost:** For additional classification models.
- **Unidecode:** To convert player names with special characters to standard English.
- **Fuzzywuzzy:** To correct and match player names, especially when dealing with decoding issues.

Chapter 5

Data Preparation

5.1 Merging and Integration

Data merging and integration strategy is critical in ensuring that data from both sources can be effectively combined to yield meaningful insights. This section outlines our methodology for merging datasets, explicitly focusing on integrating Tranfermrkt player market value data with player performance statistics from footballref dataset for the 2021-2022 and 2022-2023 seasons.

5.1.1 Merging Transfermarkt Tables

- **Context**: The Transfermarkt dataset comprises two key tables: the Players Table, which contains player profile information, and the Player Valuations Table, which records the market value of players over the past decade (refer to Figure 3.1). We aim to merge these tables to focus on player profiles and market values for the 2021-2022 and 2022-2023 seasons. This merged data will then be integrated with the fbref dataset, which contains detailed player performance statistics per 90 minutes.

- **Data Extraction**: We begin by filtering the Player Valuations Table

to include only the market values from June 2021 to May 2022. This filtered data is then merged with the Players Table to produce a combined dataset containing profile information and market valuations for the 2021-2022 season.

- **Data Merging Policy**: Given that market valuations fluctuate throughout a season, the Player Valuations Table contains multiple entries per player. To handle this, we calculate the average market value for each player across the season, creating the **2021-2022 season player data merged Table**. The same procedure is applied to the 2022-2023 season, creating the **2022-2023 season player data merged Table**.

- **Artifacts Produced**: The integration process resulted in two key datasets: **2021_2022_Player_Data_Merged Table**, which includes player profile information and average market values for the 2021-2022 season, and **2022_2023_Player_Data_Merged Table**, which contains similar data for the 2022-2023 season.

5.1.2 Integrating Transfermarkt Data with fbref Dataset

- **Context**: The next step involves merging the seasonal player valuation data derived from Transfermarkt with the corresponding player statistics from the fbref dataset for the 2021-2022 and 2022-2023 seasons.

- **Data Preprocessing**: Before merging, data preprocessing is essential to ensure compatibility between the datasets. The preprocessing tasks include normalizing names, resolving character encoding issues, and aligning feature formats.

- **Data Standardization**:

  - Player names are standardized by removing or normalizing special characters (e.g., diacritics) to maintain consistency.

  - An encoding issue in the fbref dataset caused some player names to contain question marks instead of characters. These were corrected by comparing them against the Transfermarkt dataset, using similar names (with at least 80
  - Manual verification was conducted for ambiguous cases, and unsolvable entries were excluded.

- **Country and League Standardization**: Country and league names were standardized using ISO country and league codes, ensuring uniformity across datasets and facilitating seamless integration and consistent analysis.

- **Data Merging**:

  - We merged the 2021-2022 Player Data Merged Table with the corresponding player statistics for the 2021-2022 season and repeated the 2022-2023 season process.
  - The merging was based on a composite key comprising the player's full name, domestic competition, and nationality.

- **Post-Merging Data Cleaning**: After merging, the dataset was scrutinized for duplicates and inconsistencies.

  - Rows with missing values were removed, as were duplicate rows caused by minor discrepancies (e.g., slight differences in height or market value).
  - In cases where a player had multiple entries due to a mid-season transfer, these entries were merged by averaging the numerical features while retaining the latest club and league information.

- **Data Filtering**: The final dataset was filtered to include only players who had participated in a minimum of five matches, following a crite-

rion similar to Premier League medal eligibility. This heuristic ensures that the analysis is focused on active and relevant players.

- **Final Seasonal Datasets**: The integration process yielded two comprehensive datasets for the 2021-2022 and 2022-2023 seasons, respectively. Additionally, we created position-specific tables (attackers, defenders, midfielders, and goalkeepers) to facilitate more detailed analysis. While the current research focuses on the overall dataset, these position-specific tables will be made available along with the overall dataset on Kaggle as part of our research contribution.

- **Artifacts Produced**: The integration of Transfermarkt and fbref data produced the following tables: Overall Merged Dataset 21-22 Season and Overall Merged Dataset 22-23 Season for each season's comprehensive player data, and position-specific datasets including GK Merged Dataset 21-22 Season, GK Merged Dataset 22-23 Season, Defenders Merged Dataset 21-22 Season, Defenders Merged Dataset 22-23 Season, Midfielders Merged Dataset 21-22 Season, Midfielders Merged Dataset 22-23 Season, Attackers Merged Dataset 21-22 Season, and Attackers Merged Dataset 22 23 Season. These tables support detailed analysis by season and player role.

5.1.3 Market Value Categorization

- **Context**: In line with our research design, we approach the analysis as a classification task, necessitating categorizing average market values into distinct bins. This categorization process was applied separately to all datasets generated in the previous section to account for factors such as inflation and market dynamics that could influence player valuations over time.

- **Categorization Process**: Players were categorized based on their

market value into two primary groups: above-average and below-average. The original numerical feature of `average_market_price` was retained in the dataset, as it is required for the subsequent feature selection phase, where Pearson correlations with other numerical features will be calculated. The threshold for binning was established through a distribution analysis of the target variable, `average_market_price` (in euros). The data was highly skewed, with most player market values falling below the mean. This distribution pattern, illustrated in Figures 5.1 and 5.2, suggests that most players in the top five leagues who played at least five matches during their respective seasons have market values below the average. This insight highlights the importance of identifying the characteristics distinguishing above-average players from those in the below-average category.

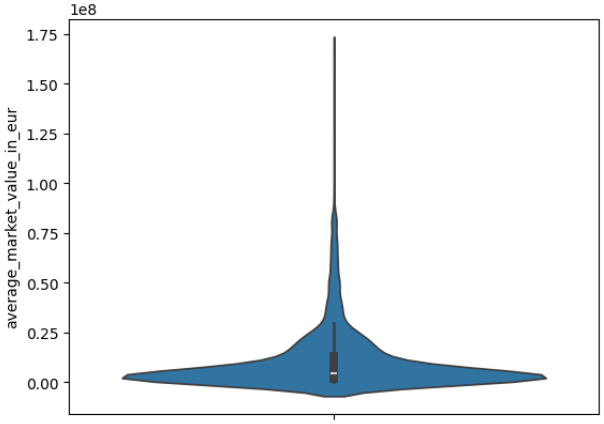


Figure 5.1: Violen plot of distribution of average markete value for 2021-2022 season

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- **Artifacts Produced**:The market value categorization was incorporated each dataset produced in the **Integrating Transfermarkt Data with fbref Dataset** second

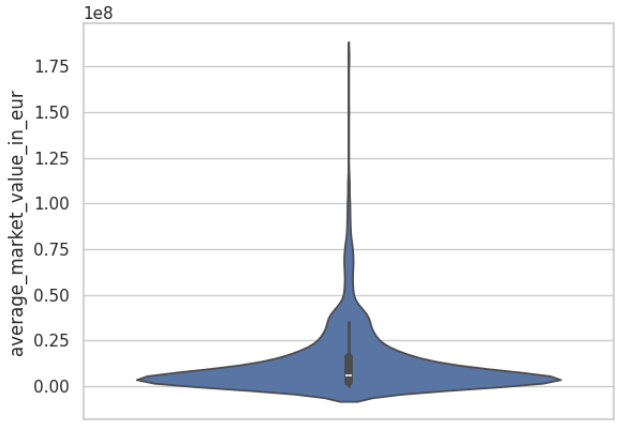


Figure 5.2: Violen plot of distribution of average markete value for 2021-2022 season

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5.1.4 Merging Data from Both Seasons

- **Context**: We merged the seasonal tables created in the previous sections to build a comprehensive dataset of player statistics and market valuations.
- **Merging Process**: The fbref dataset for the 2021-2022 season includes 143 features, while the 2022-2023 season dataset has 124 features. Of these, 115 features are common and were retained for merging.
- **Artifacts Produced**: The final datasets include **Overall Merged Dataset 21-23 Table** and position-specific tables: **GK Merged Dataset 21-23 Table**, **Defenders Merged Dataset 21-23 Table**, **Midfielders Merged Dataset 21-23 Table**, and **Attackers Merged Dataset 21-23 Table**.

Chapter 6

Feature Analysis and Selection

6.1 Introduction

This chapter provides a thorough analysis of the feature selection process before model training using the overall football player dataset we created after merging, which has 115 different player statistics.

6.2 Manual feature selection

To prevent an excessive increase in feature space, we exclude categorical features with high cardinality, as encoding them would exponentially increase the dimensionality of our dataset, preserving computational efficiency and mitigating the risk of overfitting. Hence, we exclude `Player Nationality` (with 95 unique values) and `Player club` (98 unique values).
Although `Player position` could provide context, this analysis aims to evaluate overall player performance without the influence of specific positions. Hence, the Player position is removed to ensure the analysis remains unbiased and centered on overall performance metrics. We can utilize the position-based tables we developed to perform position-specific analysis in our future work. The `Born` feature, which indicates the year of birth, is excluded as

it is redundant since we already included `Age` in the dataset, which directly captures player age information. The `Rank` feature, which represents an arbitrary unique ID assigned to each player in the Football Ref dataset, was also excluded.
Finally, `Goals` and `Assists` are straightforward indicators of high market value. Excluding them compels the model to identify less direct, more insightful indicators of market value.

6.3 Dropping Sparse Features

Features with predominantly zero values are eliminated since model algorithms might behave in unknown ways if the features have sparse data. This step reduced the dataset to 95 features, making the analysis more focused and efficient by concentrating on more informative variables. Specifically, features were dropped if 80% or more of their values were zeros. Table 6.1 shows the list of dropped features.
The dropped attributes were primarily related to set pieces, dead-ball situations, second yellow cards, red cards, and own goals. Their sparsity is logical, given that only one player among the playing eleven from a club squad of 30 players on the field typically takes set pieces and dead-ball situations. Additionally, the events involving second yellow cards, red cards, own goals, and penalty kicks are relatively rare and thus contribute to data sparsity. These occurrences are not as frequent as other player actions, leading to less representative and less consistent data for modeling.

6.4 Running Pearson Correlation

We apply a Pearson correlation threshold of 0.09 to retain features with a meaningful relationship to market value while excluding those that do not contribute significantly. This threshold balances retaining subtle yet infor-

mative features and reduces dimensionality. Moreover, it facilitates avoiding noise and overfitting. This approach ensures that weaker but potentially valuable correlations are not overlooked. This method is standard practice backed by empirical evidence and expert insights [20].

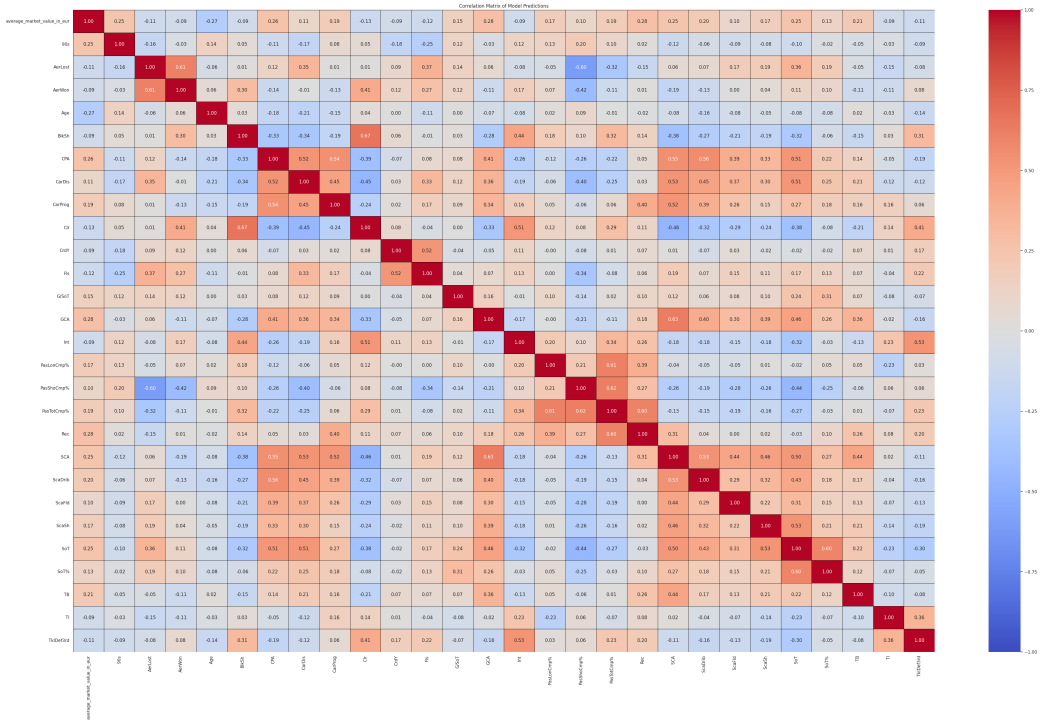


Figure 6.1: Correlation heatmap between performance metrics and the average market price of players in Europe's top five leagues.

It is important to note that Pearson correlation captures only linear relationships, so setting this threshold acknowledges that some nonlinear relationships may be missed. However, this analysis focuses on identifying linear associations, making the 0.09 threshold appropriate.

Next, to address multicollinearity, we eliminate highly correlated features (greater than 0.7) with each other. High correlations between features can destabilize the model and complicate interpretation. Removing such features ensures that each retained feature provides unique information, enhancing

the model's stability and interpretability.
Figure 6.1 presents a correlation analysis between various performance metrics and the average market price of players in EUR. The analysis highlights the top features significantly associated with higher market values, offering insights into the key attributes influencing a player's market worth.

Key Correlated Features

- **Goal Creating Actions (GCA):** Correlation: 0.2797
  Strongest positive correlation, reflecting the higher value of players contributing significantly to scoring opportunities.
- **Pass Receptions (Rec):** Correlation: 0.2779
  High correlation indicates that effective pass reception, crucial for maintaining possession and team play, enhances player value.
- **Age:** Correlation: -0.2678
  Negative correlation shows younger players tend to have higher market values, reflecting their growth potential.
- **Carries into the 18-Yard Box (CPA):** Correlation: 0.2645
  Frequent entries into the box correlate positively with market value, highlighting offensive impact.
- **Shots on Target (SoT):** Correlation: 0.2502
  Positive correlation suggests players who consistently challenge goalkeepers are more valued.
- **Shot Creating Actions (SCA):** Correlation: 0.2489
  Indicates the importance of players who facilitate scoring opportunities.
- **Minutes Played per 90 (90s):** Correlation: 0.2483
  Correlates with reliability and fitness, reflecting higher market value for consistent players.

- **Passes Behind Defenders (TB):** Correlation: 0.2137
  Positive correlation emphasizes the value of precise passes into open spaces.
- **Dribbles Leading to Shots (ScaDrib):** Correlation: 0.2015
  Players adept at dribbling past defenders to create shots are valued higher.
- **Progressive Carries (CarProg):** Correlation: 0.1913
  Positive correlation reflects the importance of advancing the ball towards the opponent's goal.
- **Total Pass Completion (PasTotCmp%):** Correlation: 0.1883
  High pass accuracy is linked to maintaining possession and supporting play.
- **Long Pass Completion (PasLonCmp%):** Correlation: 0.1741
  Emphasizes the value of precision in strategic long passes.
- **Shots Created for Further Shots (ScaSh):** Correlation: 0.1661
  Highlights the value of players who generate multiple scoring chances.
- **Goals per Shot on Target (G/SoT):** Correlation: 0.1511
  Reflects a player's efficiency in converting shots on target into goals.

6.5 Running Chi-Squared Test for Categorical Feature Importances

The chi-squared test evaluates the significance of associations between categorical variables, producing a p-value that reflects the strength of these associations. A p-value below 0.05 indicates a statistically significant relationship, whereas a p-value above 0.05 suggests no significant association.

We applied the chi-squared test to analyze the relationship between categorical features—such as player domestic competition and dominant foot—and the target variable, market value. The results are depicted in Figure 6.2.

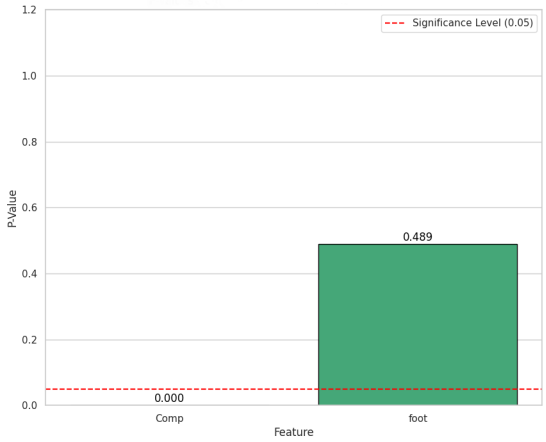


Figure 6.2: Chi Squared test p-value against features

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The analysis revealed that the p-value for the player's dominant foot is 0.489, indicating no significant relationship with market value. Conversely, the p-value for player domestic competition is close to 0, signifying a significant association. Consequently, we have excluded the dominant foot feature from our dataset.

Feature	Description
2CrdY	The number of second yellow cards a player receives, resulting in a red card.
CkStr	The number of straight corner kicks taken by a player.
CrdR	The number of red cards received by a player.
GcaDef	Goal-creating actions from defensive efforts, such as clearances or tackles leading directly to goals.
GcaDrib	Goal-creating actions resulting from successful dribbles that lead to goals.
GcaFld	Goal-creating actions resulting from fouls drawn that lead to goals.
GcaPassDead	Goal-creating actions from completed dead-ball passes (e.g., free kicks) that lead to goals.
GcaSh	Goal-creating actions from shots that lead to another goal-scoring shot.
OG	Own goals scored by a player against their team.
PKatt	penalty kicks attempted by a player.
PKcon	Penalty kicks conceded by a player, indicating how many penalty kicks the player has given away.
PKwon	Penalty kicks won by a player, representing how many penalty kicks the player has earned for their team.
ShoFK	Shots taken by a player from free kicks.
ShoPK	Penalty kicks taken by a player.

Table 6.1: Dropped sparse features

Chapter 7

Model Training

7.1 Data Preprocessing and final dataset creation

We standardize the numerical features to ensure that they contribute equally to model training, especially for algorithms sensitive to input data's scale. Standardization enhances the performance of specific models, such as K-Nearest Neighbors (KNN) and logistic regression, by placing all features on a comparable scale. This process facilitates faster convergence and improves overall model accuracy.

We applied one-hot encoding for the categorical feature: the player's domestic competition. The `Player Domestic Competition` feature, which has five unique values corresponding to Europe's top five leagues, was transformed using this method to allow the model to utilize this categorical information effectively.

The final dataset comprises both numerical and categorical features. Table 7.1 presents both the numerical features, and the categorical features.

adjustbox

Feature	Description
Numerical Features	
90s	Minutes played divided by 90. Represents the number of full 90-minute matches a player has effectively played.
AerLost	Aerial duels lost. The number of times a player has lost an aerial duel.
AerWon	Aerial duels won. The number of times a player has won an aerial duel.
Age	The player's age at the time of the match or event.
BlkSh	Number of times blocking a shot by standing in its path.
CPA	Carries into the 18-yard box. The number of times a player carries the ball into the opponent's penalty area.
CarDis	The number of times a player loses control of the ball after being tackled by an opposing player.
CarProg	Carries that move the ball towards the opponent's goal at least 5 yards, or any carry into the penalty area.
Clr	Clearances. The number of times a player clears the ball from the defensive area.
CrdY	Yellow cards. The number of yellow cards received by the player.
Fls	Fouls committed. The number of fouls a player has committed.
G/SoT	Goals per shot on target (excluding penalty kicks). Represents how many goals a player scores for every shot on target they take.
GCA	Goal-creating actions. The number of actions that directly result in a goal, including passes, dribbles, shots, fouls drawn, and defensive actions.
Int	Interceptions. The number of times a player intercepts a pass from the opposing team.
PasLonCmp%	Pass completion percentage for passes longer than 30 yards.
PasShoCmp%	Pass completion percentage for passes between 5 and 15 yards.
PasTotCmp%	Pass completion percentage for total passes.
Rec	Number of times a player successfully received a pass.
SCA	Shot-creating actions. The number of actions directly result in a shot, including passes, dribbles, shots, fouls, and defensive actions.
ScaDrib	Successful dribbles that lead to a shot attempt.
ScaFld	Fouls drawn that lead to a shot attempt.
ScaSh	Shots that lead to another shot attempt.
SoT	Shots on target (excluding penalty kicks).
SoT%	Shots on target percentage (excluding penalty kicks).
TB	Completed pass sent between back defenders into open space (through balls).
TI	Throw-ins taken.
TklDef3rd	Tackles made in the defensive 1/3 of the pitch.
Categorical Features	
Comp_bundesliga	Indicates whether the player competes in the Bundesliga (Germany's top-tier football league).
Comp_la_liga	Indicates whether the player competes in La Liga (Spain's top-tier football league).
Comp_ligue_1	Indicates whether the player competes in Ligue 1 (France's top-tier football league).

Feature	Description
Comp_premier_league	Indicates whether the player competes in the Premier League (England's top-tier football league).
Comp_serie_a	Indicates whether the player competes in Serie A (Italy's top-tier football league).

7.2 Model training and results

Stratified Train-Test Split

The average market value has been categorized into two classes: *Above Average Market Value* and *Below Average Market Value*. The distribution of the market value variable is as follows:

- **Below Average**: 2,284 examples
- **Above Average**: 858 examples

To address class imbalance during training and ensure the model generalizes well across both classes, we perform a stratified split of the data. This preserves the proportion of each class in both the training and testing sets, helping the model perform consistently across both the majority and minority classes. After the train-test split, the training set consists of 2,513 examples (1,827 below average and 686 above average), while the test set contains 629 examples (457 below average and 172 above average).

Model training using Grid Search

We train 10 distinct machine learning models, optimizing their parameters to enhance accuracy and generalization. Table 7.2 shows the list of machine learning models and their brief descriptions. We train these models individually using grid search hyperparameter tuning with stratified K-fold

Model	Description
RandomForestClassifier	Ensemble model that builds multiple decision trees and merges them for better accuracy and stability.
KNeighborsClassifier	Instance-based learning algorithm classifies data points based on the majority class of their nearest neighbors.
GradientBoostingClassifier	Boosting ensemble technique that builds trees sequentially to correct the errors of the previous trees.
GaussianNB	Probabilistic classifier based on applying Bayes' theorem with strong independence assumptions between features.
LogisticRegression	Linear model that estimates the probability of a binary outcome using the logistic function.
SVC	A linear classifier that finds the optimal hyperplane which maximizes the margin between classes in high-dimensional space.
AdaBoostClassifier	Boosting algorithm that combines multiple weak classifiers to create a strong classifier by adjusting the weights of misclassified samples.
XGBClassifier	An optimized gradient boosting ensemble framework that uses decision trees and is highly efficient for large datasets.
CatBoostClassifier	A gradient boosting ensemble algorithm that handles categorical features internally and is robust to overfitting.
HistGradientBoostingClassifier	A histogram-based boosting ensemble variant of gradient boosting that is faster and more efficient on large datasets.

Table 7.2: Model Descriptions

Model	Optimal Hyperparameters
RandomForestClassifier	criterion=entropy, min_samples_leaf=2
KNeighborsClassifier	metric=manhattan, n_neighbors=9
GradientBoostingClassifier	learning_rate=0.2, max_depth=7, min_samples_split=3
GaussianNB	Default Parameters in sklearn
LogisticRegression	C=1, solver=liblinear
SVC	C=1, class_weight=balanced, probability=True
AdaBoostClassifier	n_estimators=200
XGBClassifier	learning_rate=0.1, max_depth=7, n_estimators=200
CatBoostClassifier	learning_rate=0.2, depth=6
HistGradientBoostingClassifier	learning_rate=0.2, max_depth=7

Table 7.3: Optimal Hyperparameters for each Model

cross-validation, employing 10 folds to thoroughly explore the hyperparameter space and the optimal hyperparameters for each model is shown in Table 7.3.

Given the notable class imbalance in our target variable distribution, we use the F1 macro score as our primary metric for model selection. The F1 macro score is instrumental in this context because it provides a balanced assessment of model performance across all classes. Unlike the F1 score, which is computed for each class and then averaged, the F1 macro score calculates the F1 score for each class individually. Then it averages these scores without considering class imbalance. This approach ensures that the model's performance on minority classes is not overshadowed by the majority class, which is critical in imbalanced datasets.

Model	**CV F1 Macro**	**CV ROC AUC**	**CV F1 Micro**	**CV Accuracy**
HistGradientBoostingClassifier	0.8228	0.9111	0.8571	0.8571
XGBClassifier	0.8171	0.9090	0.8524	0.8524
CatBoostClassifier	0.8155	0.9150	0.8607	0.8607
AdaBoostClassifier	0.8119	0.8943	0.8571	0.8571
LogisticRegression	0.8096	0.9053	0.8563	0.8563
GradientBoostingClassifier	0.8086	0.8989	0.8468	0.8468
RandomForestClassifier	0.8015	0.9042	0.8532	0.8532
SVC	0.8001	0.9071	0.8281	0.8281
KNeighborsClassifier	0.7521	0.8512	0.8177	0.8177
GaussianNB	0.7274	0.8264	0.7744	0.7744

Table 7.4: Model Performance Comparison on cross-validation set

Model	**Test F1 Macro**	**Test ROC AUC**	**Test F1 Micro**	**Test Accuracy**
SVC	0.8239	0.9199	0.8521	0.8521
GradientBoostingClassifier	0.7938	0.9072	0.8442	0.8442
AdaBoostClassifier	0.7938	0.9085	0.8394	0.8394
HistGradientBoostingClassifier	0.7872	0.9038	0.8362	0.8362
XGBClassifier	0.7863	0.9035	0.8378	0.8378
CatBoostClassifier	0.7861	0.9086	0.8394	0.8394
LogisticRegression	0.7795	0.9091	0.8331	0.8331
RandomForestClassifier	0.7768	0.9033	0.8378	0.8378
KNeighborsClassifier	0.7271	0.8583	0.8013	0.8013
GaussianNB	0.7240	0.8166	0.7727	0.7727

Table 7.5: Model Performance Comparison on test set

Evaluation of machine learning models

Table 7.5 shows the results on the cross-validation set and Table 7.4 shows the test set for each model. Support Vector Classifier (SVC) outperforms other models on the test data with the highest Test F1 Macro score of 0.8239, underscoring its robustness in handling the class imbalance in the dataset. Although models such as HistGradientBoostingClassifier and XGBClassifier exhibit superior performance in cross-validation, with Cross-validation (CV) F1 Macro scores of 0.8228 and 0.8171 respectively, their slightly lower performance on the test set suggests potential overfitting or variance in model generalization. Boosting models, particularly CatBoostClassifier and AdaBoostClassifier, demonstrate consistent performance across cross-validation and test metrics, reflecting their efficacy in class-imbalanced settings. Conversely, the KNeighborsClassifier and GaussianNB models underperform, with Test F1 Macro scores of 0.7271 and 0.7240 respectively, indicating significant limitations in their ability to generalize with imbalanced classes. The SVC's superior Test F1 Macro score, in conjunction with competitive accuracy and AUC scores, positions it as the most reliable model for this dataset, aligning well with the research objective of optimizing for the F1 score due to the class imbalance challenge.

Chapter 8

Feature Importance Analysis

8.1 Feature Importance Analysis for Boosting and Ensemble Models

Several features consistently emerge as critical across the various machine learning models evaluated, while others have minimal impact. Boosting and Ensemble models demonstrate similar trends in feature importance, as depicted in Figure 8.1. Below is a summary of feature importance across these models that provide such metrics:

8.1.1 Highly Important Features

- **Age**: Frequently identified as a significant factor across models, particularly prominent in `GradientBoostingClassifier`, `CatBoostClassifier`, and `RandomForestClassifier`.
- **Comp_premier_league**: This feature is notably influential in `RandomForestClassifier`, `XGBClassifier`, and `CatBoostClassifier`, indicating that participation in the Premier League is a strong determinant in predictions.
- **90s**: Consistently critical across models, reflecting the impact of a

player's performance or presence in the 1990s. This feature holds considerable weight in `GradientBoostingClassifier`, `AdaBoostClassifier`, and `CatBoostClassifier`.

- **Rec**: Recent performance metrics are valued across all models, especially in `GradientBoostingClassifier` and `CatBoostClassifier`.
- **G/SoT and GCA**: Metrics related to goal efficiency and goal creation assists are significant across several models, particularly in `XGBClassifier` and `RandomForestClassifier`.
- **SoT**: Shots on target are a recurrently valued feature, seen as significant in `GradientBoostingClassifier`, `AdaBoostClassifier`, and `CatBoostClassifier`.

8.1.2 Moderately Important Features

- **PasTotCmp%**: Overall passing completion percentage holds moderate importance across multiple models, including `RandomForestClassifier` and `AdaBoostClassifier`.
- **TklDef3rd**: Tackles in the defensive third show moderate significance, particularly in `AdaBoostClassifier`.
- **Comp_ligue_1**: Presence in Ligue 1 is of moderate importance in `XGBClassifier` and `CatBoostClassifier`.
- **SCA**: Successful actions or dribbles are important in `CatBoostClassifier` and `XGBClassifier`.

8.1.3 Least Important Features

- **Comp_bundesliga**: Participation in the Bundesliga is often considered of low importance across models.

- **AerLost and Int**: Defensive metrics such as aerial duels lost and interceptions show less consistent importance across different classifiers.

Overall, the most important features identified by these models include **Age**, **Comp_premier_league**, **90s** and **Rec**, with a particular emphasis on recent performance metrics and goal-related statistics. Features related to specific league participation (aside from the Premier League) typically have minimal influence on model predictions.

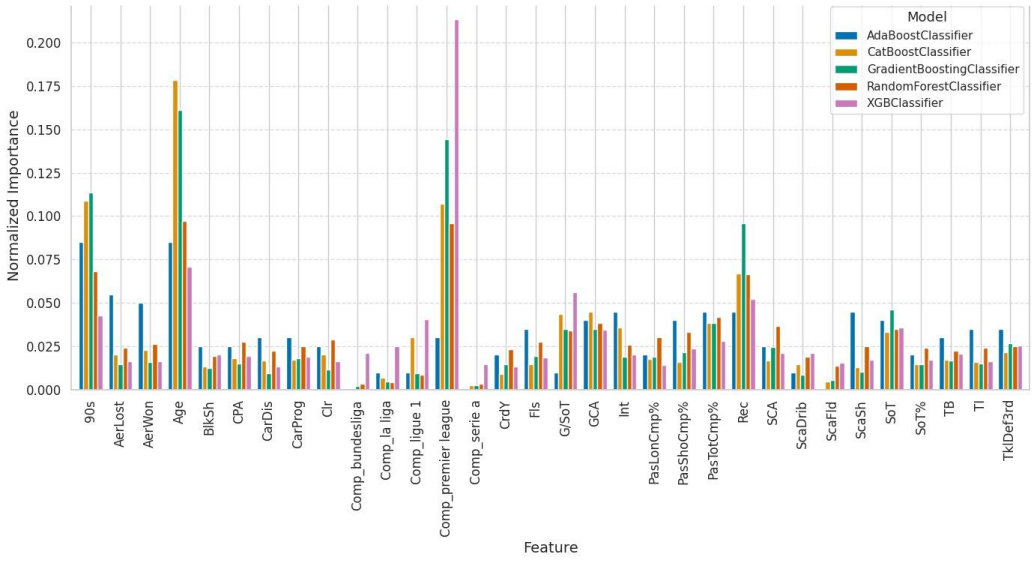


Figure 8.1: Clustered bar chart of normalized feature importances for Boosting and ensemble models

8.1.4 Feature Importance for logistic regression model

Figure 8.2 plots the logistic regression coefficients with the relative importance of each feature in predicting the target outcome. The most influential feature is participation in the "Premier League" (`Comp_premier_league`), with a coefficient of 1.8896, indicating a strong positive impact on the target

outcome. Significant positive contributions also come from "90s" (coefficient: 1.1953), representing playing time, and "Shots on Target" (`SoT`, coefficient: 0.7614), highlighting offensive effectiveness. In contrast, "Comp_ligue_1" (coefficient: -1.7141) and "Age" (coefficient: -1.3724) have substantial negative associations, suggesting that players in Ligue 1 or older players are less favorable. Features with coefficients near zero, such as `ScaFld` (-0.0355) and `ScaDrib` (-0.0258), have minimal impact, indicating they do not significantly influence the model. These results underscore the importance of league competition and player experience, with the Premier League and younger players being more predictive of positive outcomes, which is crucial given the class imbalance in the dataset.

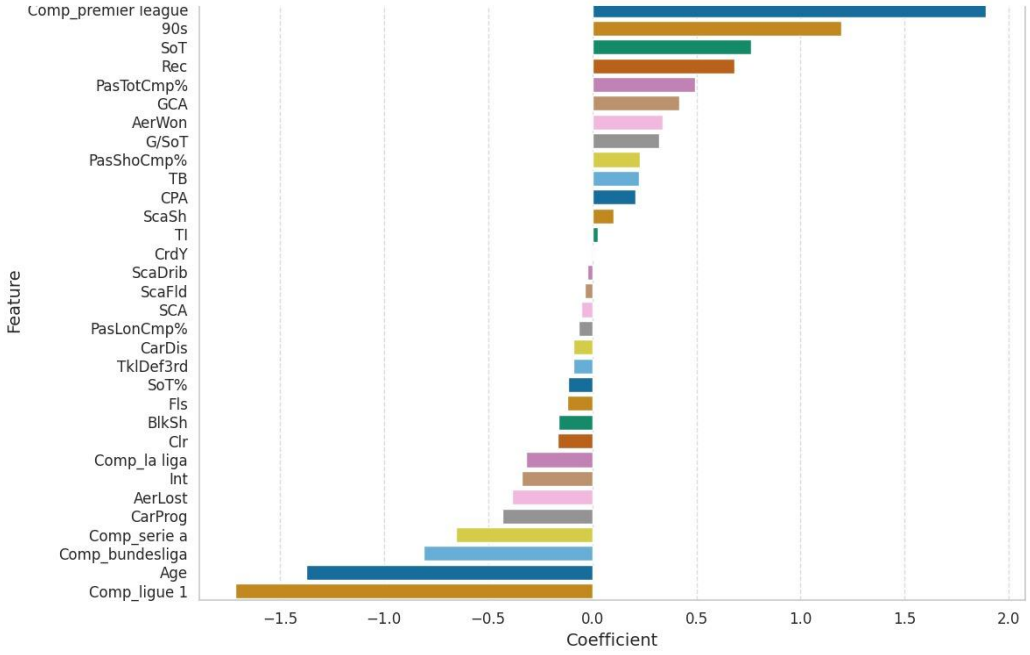


Figure 8.2: Logistic regression coefficients illustrating the relative importance of each feature

8.1.5 Comparison of Feature Importance Across Models

They compare feature importance across the boosting/ensembled models and the logistic regression analysis reostly consistent insights. Features like `Comp_premier_league` and `90s` are essential in both analyses, with high positive coefficients in logistic regression reflecting their strong influence on predictions and consistently high importance across other models. `Age` shows a significant negative coefficient in logistic regression, indicating high importance but a negative impact. In contrast, feature importance models highlight it as a critical determinant. `Comp_ligue_1` is considered moderately important in feature importance models and shows a strong negative coefficient in logistic regression, marking it as less favorable in predicting positive outcomes. Features such as `ScaFld` and `ScaDrib`, with near-zero coefficients in logistic regression, align with their low importance across feature importance models, indicating consistent insignificance. In contrast, logistic regression coefficients provide significant weights to Comp_ligue_1 Comp_bundesliga and Comp_seria_a compared to boosting and ensemble models.

Chapter 9

Voting Classifier

9.0.1 Model selection and training

Our analysis of feature importance demonstrated that boosting and ensemble models consistently attribute similar levels of importance to features. Hence, we proceeded with a final experiment involving the design of a soft margin voting classifier by integrating multiple boosting and ensemble models. This model, referred to as the *Combined-ensembled Model*, was developed to address the potential issue of boosting and ensemble models with similar bias disproportionately influencing the final predictions if we were to design a combined voting classifier of all ten machine learning models. Moreover, we excluded KNN and GaussianNB due to their poor performance, as evident by their CV and test scores shown in Figure 8.1 and Figure 8.2 above. We trained a soft margin voting classifier using a Support Vector Classifier (SVC), Logistic Regression, and the Combined-ensembled model voting classifier to create a less biased and more accurate model.

9.0.2 Model training and results

The model was trained and evaluated on the stratified train test split, and the results are depicted in Table 9.1. For the test set, this voting classifier

achieved the highest test accuracy, ROC AUC score, and F1_micro score compared to all individual models. Although the test set F1_macro score was slightly lower than that of the best-performing individual model (SVC with 0.823), it was relatively close with 0.804. More importantly, regarding cross-validation F1 macro scores, the voting classifier outperformed all other trained models, which is a more significant metric as it demonstrates superior generalization and stability across different subsets of the data, providing a more reliable estimate of model performance.

Metric	Result
Voting Classifier Test Accuracy	0.8553
Voting Classifier F1 Score (Weighted)	0.8498
Voting Classifier F1 Score (Macro)	0.8048
Voting Classifier F1 Score (Micro)	0.8553
Voting Classifier ROC AUC Score	0.9223
Mean Cross-Validated F1 Macro Score	0.8265

Table 9.1: Voting Classifier Model Performance

Chapter 10

Conclusion, Limitations, and Recommendations

10.1 Conclusion

While the literature on football player market valuation offers a range of innovative methodologies, it has substantial limitations. The prevailing models are predominantly regression-based and constrained by a narrow scope of performance metrics, which often fail to encapsulate the multifaceted nature of the football transfer market. These conventional approaches exhibit considerable shortcomings in accommodating the inherent volatility, noise, and outlier influences within market data, frequently resulting in predictions that lack precision and practical utility.

This dissertation seeks to bridge these gaps by proposing a classification-based framework for predicting player market values, diverging from focusing on regression techniques. This approach simplifies the decision-making process by categorizing market values rather than estimating exact numerical figures while enhancing interpretability—rendering it practically advantageous for football clubs and scouts. Deploying classification models confers greater efficiency against market data's volatility and stochastic disturbances, yield-

ing more stable and reliable predictive outcomes. Such a methodology augments model performance and scalability and holds the potential for broader applicability in other sectors where asset valuation is paramount.
Moreover, this research introduces an innovative voting classifier, meticulously designed by scrutinizing and addressing biases in feature importance, thereby refining the model to achieve higher accuracy and fairness. Furthermore, our examination of correlations and feature importance identified critical variables influencing player market values and their associated weights in influencing the market values.
Furthermore, this study meticulously curates a novel dataset synthesized from two highly reputable sources spanning two seasons across Europe's top five leagues. This dataset is synthesized and customised from two reputable sources each to maximise data quality and breadth. If allowed after the review of this project we hope to make dataset publicly available on Kaggle and other platforms to encourage further exploration and research in this area.

10.2 Limitations and Future work

In our dissertation, although we developed a player-specific dataset, its detailed analysis was beyond the scope of this study. For future research, we intend to explore these datasets further and conduct an analysis similar to that of the overall player dataset across different player positions. Given that the underlying variables remain consistent, this approach shall be feasible.
Additionally, our findings revealed that a player's participation in certain leagues, such as the Premier League, highly influences their market values. This observation suggests a need for league-specific analyses to understand how valuations fluctuate across competitive contexts. Future work will focus on assessing player valuations in relation to specific leagues, which may uncover patterns and insights unique to each league's dynamics.

Lastly, the dataset we curated encompasses player data spanning only two seasons. We plan to expand this temporal scope by acquiring historical data from the FBref website, contingent upon compliance with data access guidelines. If permitted, we will extract and analyze data from prior seasons, which may enable us to refine our model and enhance its ability to predict player market value accurately so that it can be deployed into a real-world setting.

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